

Swell Audit Summary

Prepared by Cyfrin

Version 2.0

Lead Auditors

Dacian

Carlitox477

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1 About Cyfrin

Cyfrin is a Web3 security company dedicated to bringing industry-leading protection and education to our partners and their projects. Our goal is to create a safe, reliable, and transparent environment for everyone in Web3 and DeFi. Learn more about us at cyfrin.io.

2 About This Document

This document is a public summary of the security audit Cyfrin performed for Swell on Swell. It records the repositories and commits reviewed, the timeline, auditors, in-scope files, and the number and final status of findings by severity. It closes with Cyfrin's attestation of the review's scope and final remediation status.

It is deliberately abbreviated: the technical detail of individual findings is not reproduced here. The complete audit report, version 2.0 dated February 22, 2024, was delivered to Swell, who retain it as the full record of the engagement. This summary was issued on October 29, 2026.

3 Disclaimer

The Cyfrin team makes every effort to find as many vulnerabilities in the code as possible in the given time but holds no responsibility for the findings in this document. A security audit by the team does not endorse the underlying business or product. The audit was time-boxed and the review of the code was solely on the security aspects of the in-scope code being audited. Unless the scope explicitly notes otherwise, the review is limited to the application layer and does not extend to the correctness of API/SDK/execution environment implementations, compilers, standard and third-party libraries, underlying operating systems and hardware.

4 Copyright and License

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5 Audit Overview

Over the course of 18 days, the Cyfrin team conducted an audit on the Swell code provided by Swell. The findings consist of 2 Medium & 11 Low severity issues with the remainder being informational and gas optimizations.

Summary

Project Name	Swell
Repository	v3-contracts-lst
Commit	a95ea7942ba8...
Audit Timeline	Jan 29th - Feb 21st, 2024
Methods	Manual Review, Stateful Fuzzing

Issues Found

Severity	Found	Resolved	Acknowledged
Critical Risk	0	0	0
High Risk	0	0	0
Medium Risk	2	2	0
Low Risk	11	8	3
Informational	4	4	0
Gas Optimizations	7	5	2
Total	24	19	5

6 Audit Scope

The Barracuda Upgrade is a significant upgrade to Swell's protocol which allows holders of swETH to withdraw by burning their tokens in exchange for ETH.

The following contracts were included in the scope for this audit:

```
contracts/implementations/AccessControlManager.sol
contracts/implementations/DepositManager.sol
contracts/implementations/NodeOperatorRegistry.sol
contracts/implementations/RepricingOracle.sol
contracts/implementations/swETH.sol
contracts/implementations/swEXIT.sol
contracts/implementations/Whitelist.sol
contracts/interfaces/IAccessControlManager.sol
contracts/interfaces/IDepositManager.sol
contracts/interfaces/INodeOperatorRegistry.sol
contracts/interfaces/IRepricingOracle.sol
contracts/interfaces/IswETH.sol
contracts/interfaces/IswEXIT.sol
contracts/interfaces/IWhitelist.sol
contracts/libraries/DepositDataRoot.sol
contracts/libraries/EnumberableSetValidatorDetails.sol
contracts/libraries/Repricing.sol
contracts/libraries/SwellLib.sol
contracts/vendors/AggregatorV3Interface.sol
contracts/vendors/IDepositContract.sol
contracts/vendors/IPorAddresses.sol
contracts/vendors/IRateProvider.sol
```

7 Risk Classification

	Impact: High	**Impact: Medium**	**Impact: Low**
Likelihood: High	Critical	High	Medium
Likelihood: Medium	High	Medium	Low
Likelihood: Low	Medium	Low	Low

8 Protocol Summary

Swell is a Liquid Staking Derivatives (LSD) protocol which allows users to participate in and benefit from Ethereum staking without having their funds locked up or running validators. Users are able to deposit their ETH into the protocol in exchange for swETH, Swell's liquid ERC20 token.

Once enough ETH has been deposited into the protocol, node operators use this ETH to run validators and participate in Ethereum's proof of stake consensus layer, thereby earning rewards. As rewards are earned, Swell re-calculates the swETH/ETH exchange rate such that holders of swETH can burn their swETH to withdraw a greater amount of ETH than they originally deposited. Operators and the Swell protocol are compensated with a percentage of the ETH rewards in swETH.

The swETH withdrawal process is handled by the swEXIT contract, an ERC-721 token minted when a user requests a withdrawal by specifying the amount of swETH to burn in exchange for ETH. Once the withdrawal request has been processed, the ETH can be claimed, which burns the NFT representing the request and sends the ETH to its owner.

9 Attestation

Cyfrin attests that it performed a security review of the Swell code provided by Swell.

The repositories, commits, review period and findings with their final status are in the Summary and Issues Found tables in Section 5, and the in-scope files in Section 6. This attestation is limited to the documented scope, audited commits and mitigations specifically reviewed by Cyfrin.

Cyfrin publishes this document and its SHA-256 hash at https://github.com/Cyfrin/cyfrin-audit-reports. To check a copy on macOS or Linux, run the following in the folder that contains it; it prints `OK` when the copy matches. If the copy was renamed, set F to its file name. Before running it, check that the URL is Cyfrin's repository.

```
U=https://github.com/Cyfrin/cyfrin-audit-reports/raw/main/reports/sha256
N=2024-02-23-cyfrin-swell-barracuda-v2.0.pdf
F=$N
H=$(curl -fsSL $U/$N.sha256 | cut -d' ' -f1)
echo "$H *$F" | shasum -a 256 -c
```

Cyfrin's full audit report with all findings provided to Swell has the SHA-256 hash and file name below. To check a copy of it, first check this document as above, then run the following in the folder that contains the full report; it prints `OK` when the copy matches. If the copy was renamed, set F to its file name.

```
H=e52156fca424b19487a9145c1dbae6cd2f164f388dada5fcc044b4cdf405cf28
F=2024-02-23-cyfrin-swell-barracuda-v2.0-full.pdf
echo "$H *$F" | shasum -a 256 -c
```

On Linux, `sha256sum -c` can be used in place of `shasum -a 256 -c`. On Windows, run `Get-FileHash <file> -Algorithm SHA256` in PowerShell and compare the result with the hash.